

Pattern Recognition

Syllabus

INT422-Pattern Recognition

Fall-2026

Sunday 9:00am-11:00am, N301

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Course Description and Prerequisites

INT422—Pattern Recognition (2+2+0) 3 credits

Official Course Description: Introduction; Probability theory: Bayes' rule; Parameter estimation; Statistical decision making; discriminate functions; measures of classification performance and measures of classification risk; Non-parametric decision making; Adaptive discriminate functions; Minimum squared error discriminate functions; Clustering techniques: Hierarchical clustering, Partitioning clustering; Artificial neural networks; Hopfield nets – Other PR systems: Syntactic pattern recognition; Hidden Markov Model based: Application examples.

Prerequisite(s): MA110—Linear Algebra & MA111—Calculus.

About the Teaching Assistants Teaching assistants can help you with the course material, including labs, assignments and exams. They will do most of the assignment marking.

Teaching Assistants:

- Aya M. Moubarak aya.mubarak@su.edu.eg

Student Learning Outcomes

After completing this course, the students should be able to:

1. Understand basic concepts in pattern recognition.
2. Gain knowledge about state-of-the-art algorithms used in pattern recognition applications.
3. Understand pattern recognition theories, such as Bayes classifier, linear discriminant analysis.
4. Prototype, model, and test various pattern recognition algorithms.
5. Apply pattern recognition techniques in practical problems.

List of Required Course Materials

Required Textbook: *Pattern Classification*, 2nd edition, by R.O. Duda, P.E. Hart & D.G. Stork, Wiley Inc., 2001. The textbook has a website: <http://www.rii.ricoh.com/~stork/DHS.html>.

Recommended Supplemental Textbook(s):

1. *Pattern Recognition: An Algorithmic Approach* by M. Narasimha Murthy & V. Susheela Devi, Springer, 2011.
2. *Pattern Recognition and Machine Learning*, 2nd edition, by C. M. Bishop, Springer, 2006.
3. *Pattern Recognition*, 4th edition, by S. Theodoridis & K. Koutroumbas, Academic Press, 2009.
4. *An Introduction to Statistical Learning*, 2nd edition, by G. James, D. Witten, T. Hastie and R. Tibshirani, Springer, 2021.
5. *Essentials of Pattern Recognition An Accessible Approach*, 1st edition, by J. WU, Cambridge University Press, 2021.
6. *Machine Learning Fundamentals A Concise Introduction*, 1st edition, by J. Jiang, Cambridge University Press, 2021.

Course Material Distribution: The course material as well as assignments and labs will be distributed mostly electronically on LMS and on the class website (if any). It is your responsibility to check for new material frequently, daily is recommended.

Topics Outline

The topics below correspond chapters in Duda & Hart, but you should only consider them a guide to what we'll go over. Tentatively, here's what we're going to cover:

Week 1	Introduction, Bayesian Decision Theory
Week 2	Maximum Likelihood, Bayesian Parameter Estimation
Week 3	Parameter Estimation, Component Analysis, and Discriminants
Week 4	Non-Parametric Techniques
Week 5	Linear Discriminant Functions
Week 6	Midterm
Week 7	Support Vector Machines and Kernel Methods
Week 8	Stochastic Methods, Graphical Models
Week 9	Neural Networks
Week 10	Deep Learning
Week 11	Unsupervised Learning and Clustering
Week 12	Time Series
Week 13	Applications
Week 14	Final Revision

Teaching Methods

The course will be based on the following teaching and learning activities:

1. **LECTURES:** covering the theoretical part using both Power-Point presentations, *online sessions* and White board.
2. **QUIZZES:** review questions and class discussion.

3. LAB SESSIONS: covering the programming assignments.
4. COURSE PROJECTS: apply the theoretical algorithms in real-world applications related to the course topics.

Approximate Schedule of Exams

There will be midterm and a comprehensive final exams. These exams will be held during the *official Sinai University exam period*. The final exam cover will be comprehensive and may consist of *multiple-choice*, *true/false*, and *programming questions*. Your responsibility for this course may include attendance at a final exam. The exact dates will depend on our progress through the material. If you are not present when an exam is given, you will receive a zero for that exam. Suggested strategy: keep current, attend class, and be ready.

Assignments, Quizzes and Labs

Homework

Will be assigned each week as Python Notebook problem sets, these are to be completed and then uploaded to Canvas for review.

Labs

Computer labs will be assigned during the term. The emphasis of these labs will be to provide insights into data science algorithms, complementing the more analytical material discussed in class. The labs will be graded based upon the submitted lab reports and lab exam. Lab reports will be submitted to the TAs by the due date.

Projects

There will be three projects applying the skills students developed to actual datasets. The first project will have students analyze a well-known dataset, the final project will have the students work on a modeling problem of their choice.

Quizzes

There will be a short (20 minute) in-class quiz each week. There are two attempts at the quiz and the questions are pulled randomly during each attempt. The higher of the two attempts counts for the grade. Your lowest quiz grade will be dropped.

Discussion Forums

There will be weekly discussion forums to give students the opportunity to read and write about the material in the course.

Journal Club

As part of this course, students will participate in Journal Club designed to develop critical reading, presentation, and discussion skills. Each student will present one peer-reviewed research articles during the semester focus on pattern recognition in real applications, especially those involving advanced or emerging techniques not covered in this course. Your goal is to explore how these methods are being applied to cutting-edge problems in real applications and to demonstrate your ability to critically analyze, interpret, and communicate complex pattern recognition-driven research.

Attendance Policy

I don't take daily attendance. Missing class with any regularity, however, will affect your course grade directly through the class participation grade and indirectly through its impact on your performance. Slides do not reflect all materials covered in lectures. Students are responsible for material covered in any class that they do not attend. Students are also expected to check this website regularly for latest announcements, upcoming events and released homework. Absence from lectures and/or tutorials shall not exceed 25%. Students who exceed the 25% limit without a medical or emergency excuse acceptable to and approved by the university council shall not be allowed to take the final examination and shall receive a mark of zero for the course. If the excuse due to a severe medical condition or exceptional circumstance approved by the university council, the student withdraws from the course and withdrawal rules applies. The student receives a **W** grade on his/her academic record.

Late Submission Policy

Homework may be submitted late by up to two days; the penalty for late submission increasing linearly from 10% to 100% of the homework score.

Makeup Policy

Unless a good reason and its supporting evidence are given - e.g. due to illness or emergency events - no makeup is acceptable for students missing a homework assignment or an exam.

Grading and Assessment

The grades will be based on the completion of quizzes, assignments, two examinations (midterm and final), and a team-based software development project. To do well in this course, you must do well on both the assignments, team-based project, *and* the quizzes. These two different forms of evaluation test different aspects of the material—how well you understand the underlying concepts (exams) and how well you can put the concepts into practice (projects and labs). There will be midterm and final exams. Your grade will be determined by your performance on the *team project*, *practical*, *assignments* and *exams*. The percentage breakdown is:

Weekly Assignment	10 %
Team Project	10 %
Midterm Exam	20 %
Final Exam	60 %

Additionally, students will be required to complete a small (but nontrivial) project related to their intended area of interest. They will have to present their work before the end of the semester. The items listed above will constitute 100% of the graduate section grade. The project will constitute 10%.

Grade Settings

If your overall grade falls within one of the prescribed ranges, then you are guaranteed to receive at least the letter grade indicated.

Grading Scale	
Score	Grade
95%–100%	A+
90%–95%	A
85%–90%	A-
80%–85%	B+
75%–80%	B
70%–75%	C+
65%–70%	C
60%–65%	D
Less than 60%	F

Class Project

INT422 will have a class project that requires students to participate in working teams where students organize, manage, and practice the pattern recognition application. Each project requires to develop a software product for an imaginary customer. Small teams of 4 students will compete to deliver solutions to a specified recognition problem. Project objectives will be discussed in class.

- *Projects will include a peer presentation at the end of the class.*
- *Your final grade for this component will include peer evaluation grades provided by your team mates.*

Team Policy

- Everyone must take part in a team project
- All members of a team will receive the same score; that is, the project is assessed and everyone receives this score.
- Everyone in team must take part in a project.
- Teams cannot be altered or switched.

Programming Environment

The course has an associated laboratory, which will require the use of Matlab or Scientific Python as recommended tools for the class. Programming examples will be shown in class using a mix of Matlab or Python based on what the instructor have available. Students are encouraged to learn and use Python (i.e., SciPy, NumPy) in the course. Many programming materials given in lecture and programming aspects of the homework will be given in Python. Also, there are several public-domain tool libraries are available for running experiments.

Statement on Academic Dishonesty

Cheating, plagiarism or otherwise obtaining grades under false pretenses constitute academic dishonesty according to the code of this university. Academic dishonesty will not be tolerated and penalties can include canceling a student's enrollment without a grade, giving an **F** for the course or for the assignment. For more details, see the *University of Sinai* rules.

Use of Artificial Intelligence (AI)

Sinai University is committed to exploring and using artificial intelligence (AI) tools as appropriate for the discipline and task undertaken. All uses of AI should be acknowledged as this aligns with our commitment to honor and integrity, as noted in Sinai University's Honor Code.

For this course, you can use AI programs (ChatGPT, Gemini, Copilot, etc.). These programs can be powerful tools for learning and other productive pursuits, including completing assignments in less time, helping you generate new ideas, or serving as a personalized learning tool. However, your ethical responsibilities as a student remain the same. You must follow Sinai University's Honor Code and uphold the highest standards of academic honesty. This applies to all uncited or improperly cited content, whether created by a human or in collaboration with an AI tool. If you use an AI tool to develop content for an assignment, you must cite the tool's contribution to your work.

Statement for Academic Success Services

Your student fees cover usage of the computer laboratories. These laboratories support your classroom learning; it is your responsibility to take advantage of their services. Keep in mind that seeking help outside of class is the sign of a responsible and successful student.

Professionalism Policy

Per university policy and classroom etiquette; mobile phones, iPods, etc. must be silenced during all classroom and lab lectures. Those not heeding this rule will be asked to leave the classroom/lab immediately so as to not disrupt the learning environment. Please arrive on time for all class meetings. Students who habitually disturb the class by talking, arriving late, etc., and have been warned may suffer a reduction in their final class grade.

Subject to Change Statement

All information contained in this syllabus is subject to change, with advance notice, as deemed appropriate by the instructor.